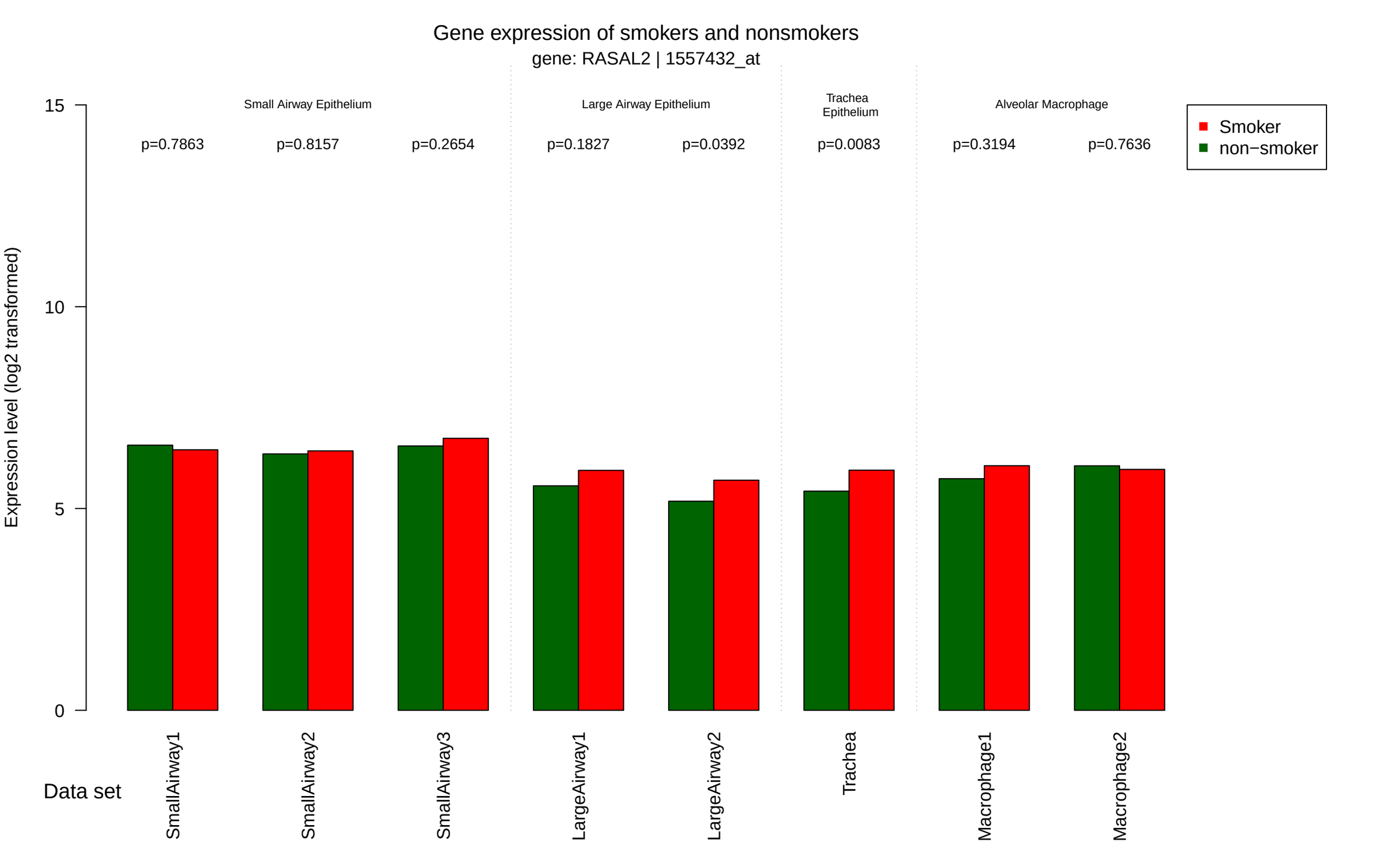
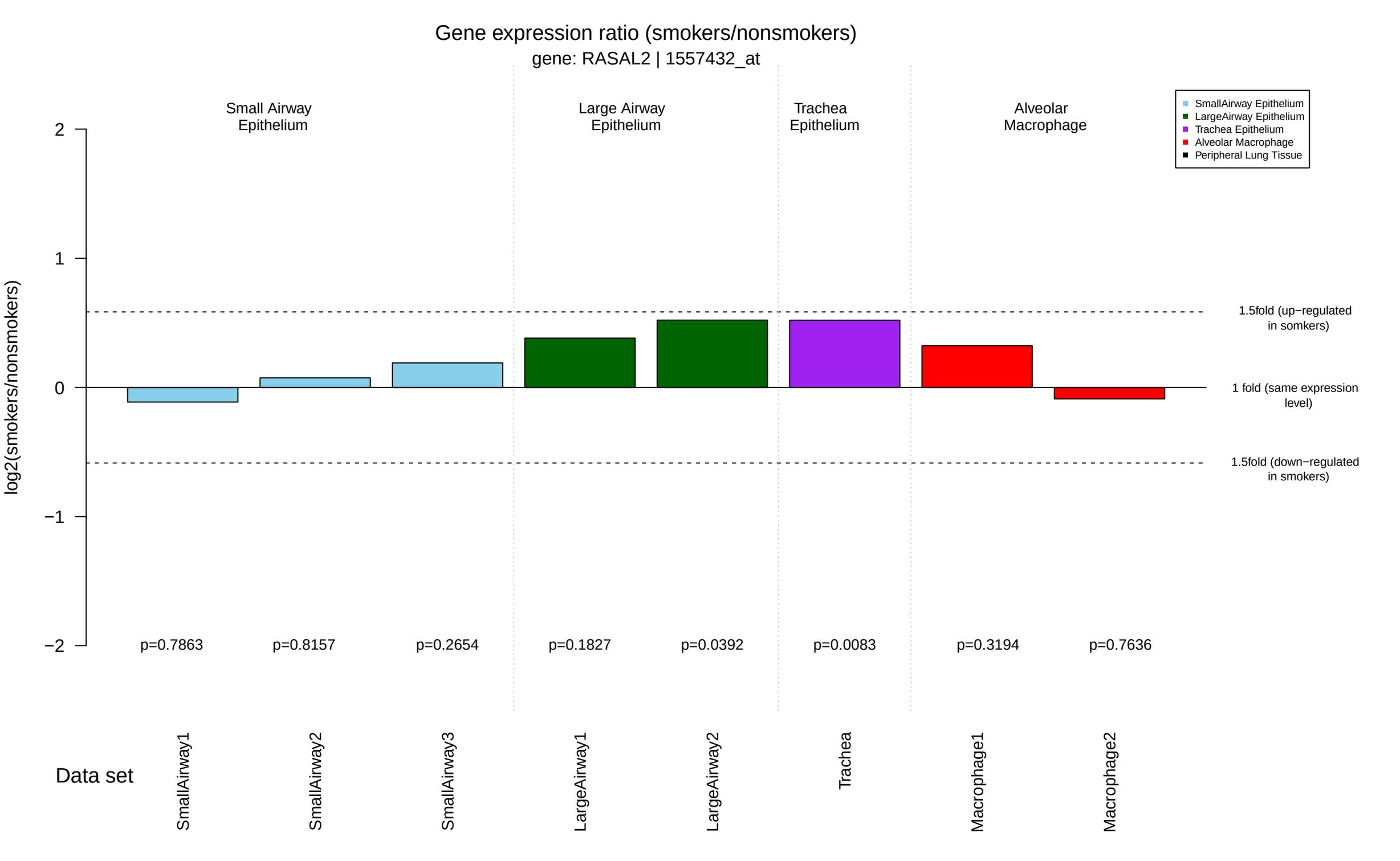
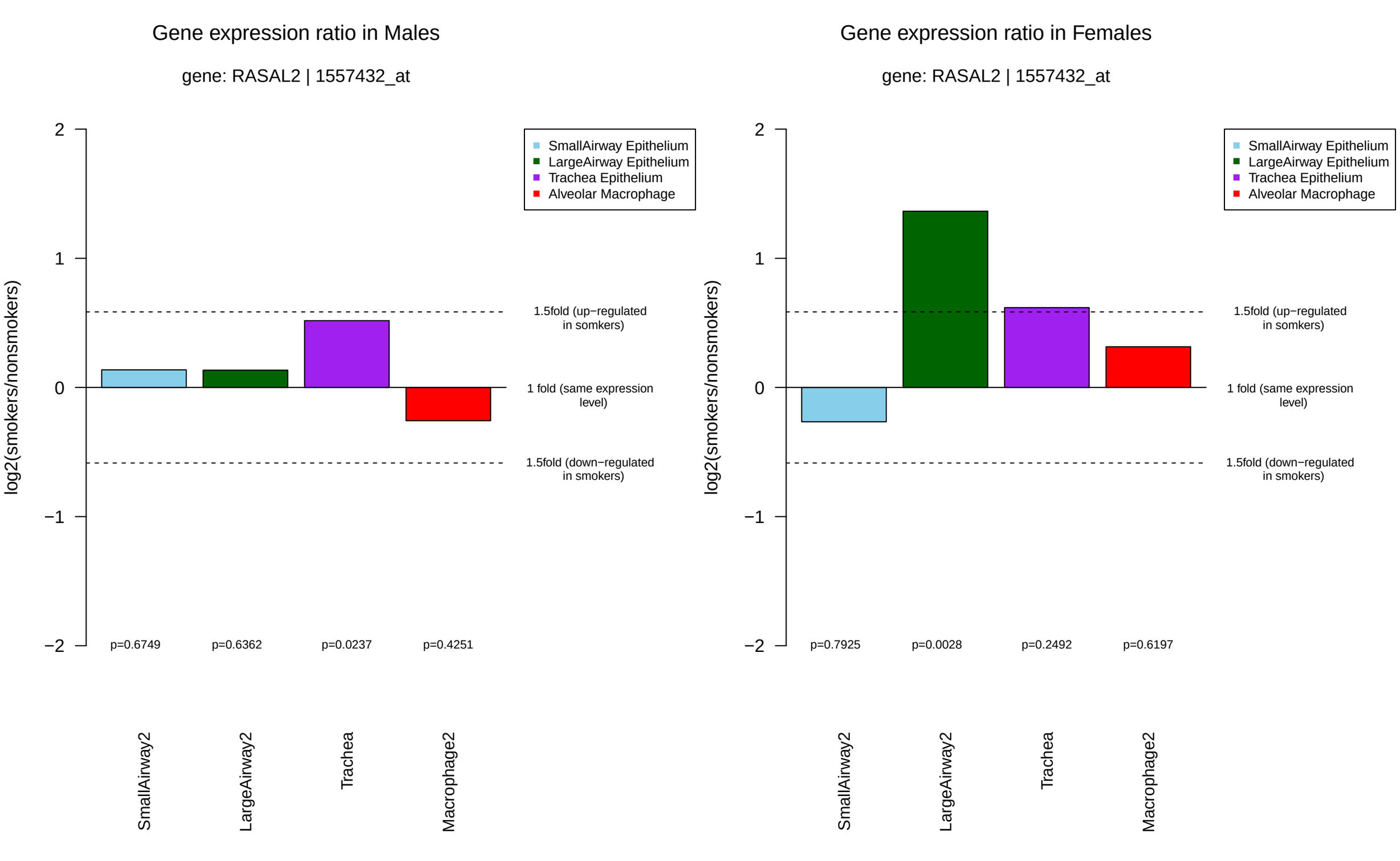
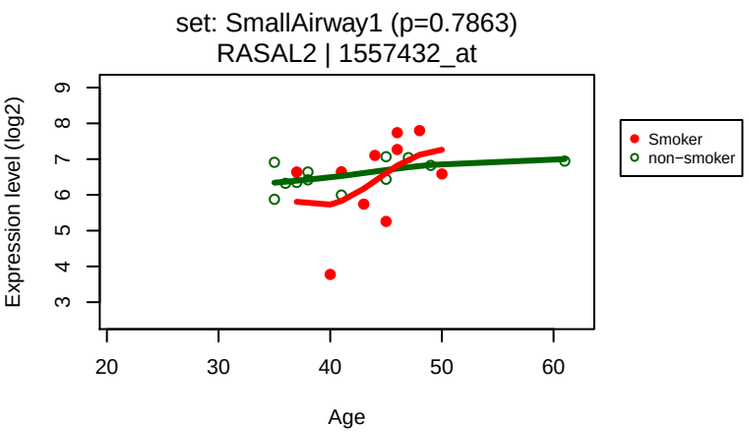
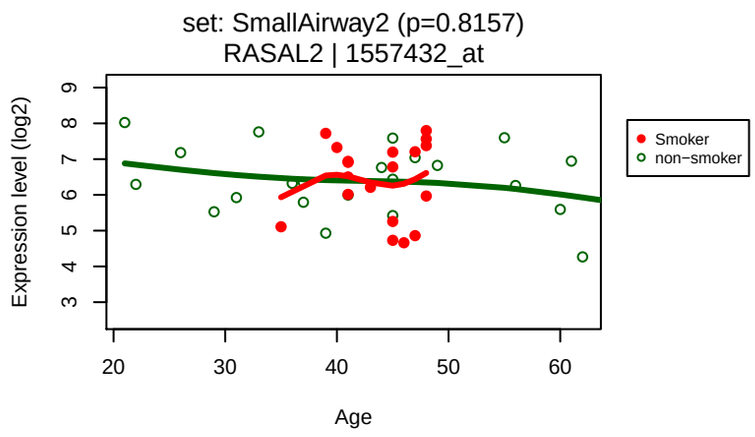
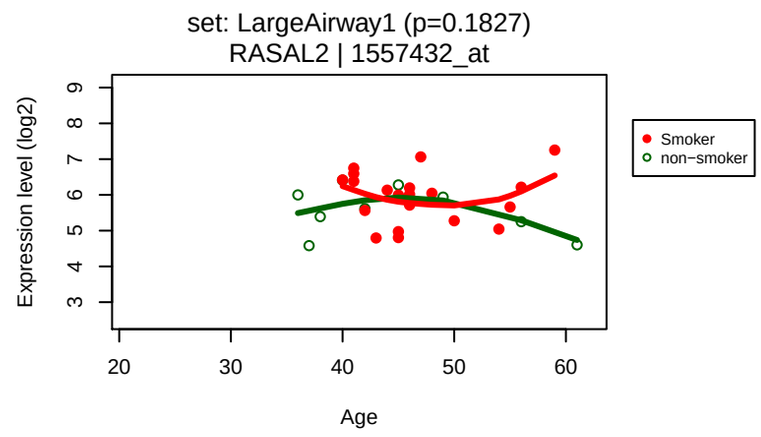

Gene expression of smokers and nonsmokers
gene: RASAL2 | 1557432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7863
p=0.8157
p=0.2654
p=0.1827
p=0.0392
p=0.0083
p=0.3194
p=0.7636
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

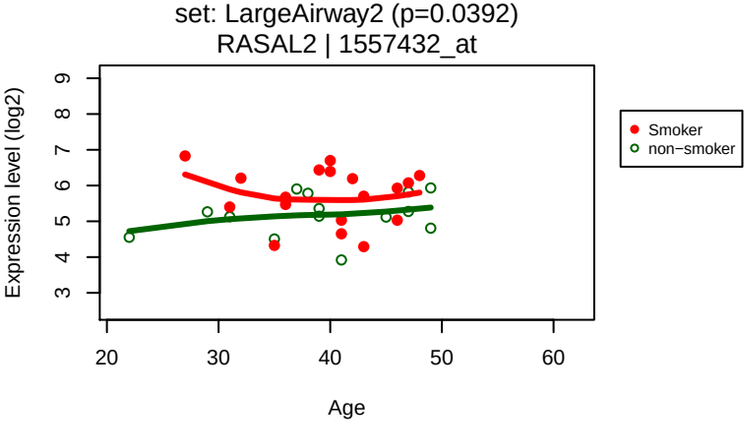

Gene expression ratio (smokers/nonsmokers)
gene: RASAL2 | 1557432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7863
p=0.8157
p=0.2654
p=0.1827
p=0.0392
p=0.0083
p=0.3194
p=0.7636
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

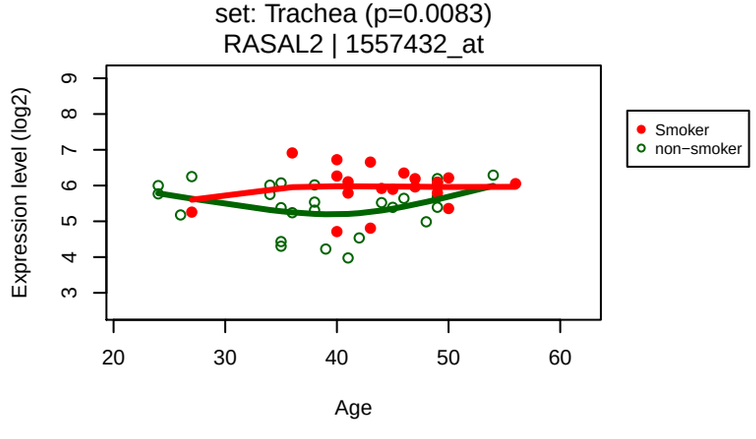

Gene expression ratio in Males
gene: RASAL2 | 1557432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6749
p=0.6362
p=0.0237
p=0.4251
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASAL2 | 1557432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7925
p=0.0028
p=0.2492
p=0.6197
SmallAirway2
LargeAirway2
Trachea
Macrophage2

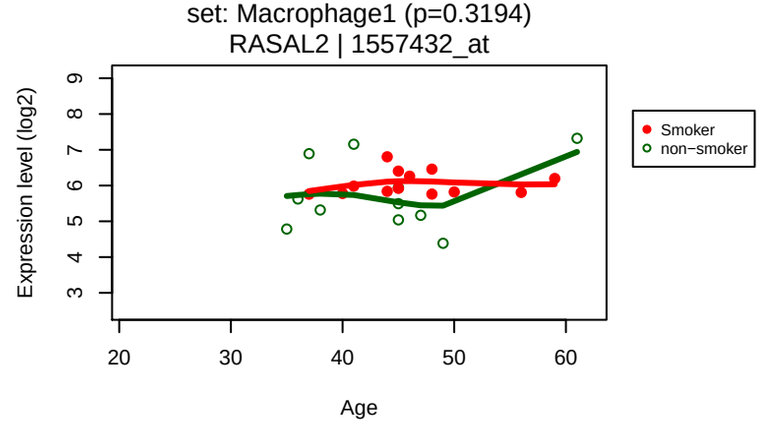

set: SmallAirway1 (p=0.7863)
RASAL2 | 1557432_at
Expression level (log2)
Age
Smoker
non−smoker

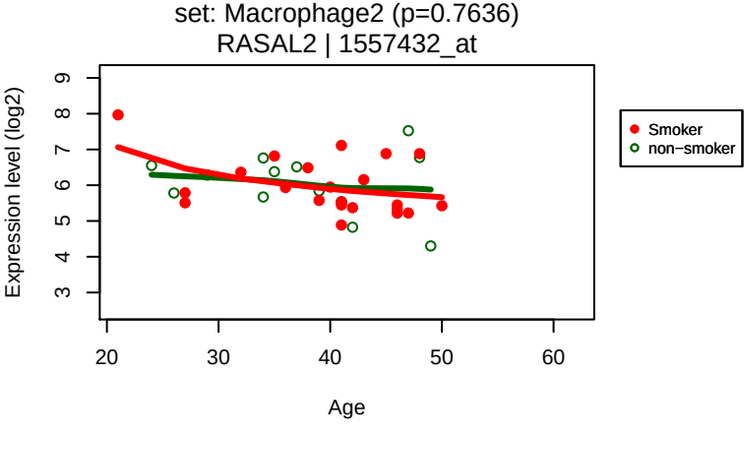

set: SmallAirway2 (p=0.8157)
RASAL2 | 1557432_at
Expression level (log2)
Age
Smoker
non−smoker

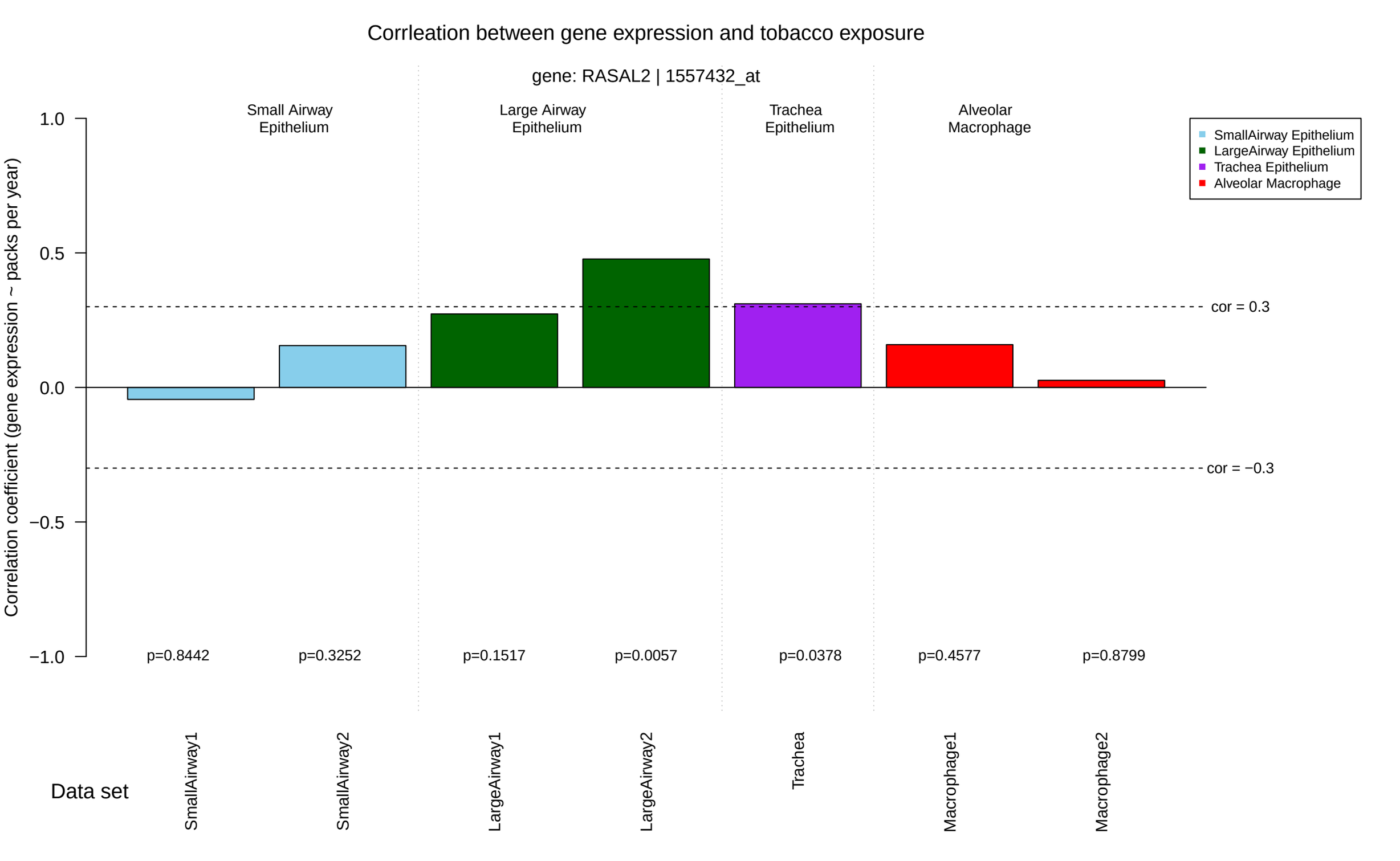

set: LargeAirway1 (p=0.1827)
RASAL2 | 1557432_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0392)
RASAL2 | 1557432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0083)
RASAL2 | 1557432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3194)
RASAL2 | 1557432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7636)
RASAL2 | 1557432_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASAL2 | 1557432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8442
p=0.3252
p=0.1517
p=0.0057
p=0.0378
p=0.4577
p=0.8799
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2